

XINYU HUANG

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EDUCATION

The University of Sydney

Sydney, AUS

Bachelor of Engineering Honours (Software Engineering)

Feb 2022 to Present

Cumulative GPA: 3.53/4.00

Relevant Courses: *Enterprise Scale Software Architecture, Machine Learning and Data Mining, Communication Networks, Introduction to Artificial Intelligence, Distributed Systems, Algorithm Design, Data and Information Management, Models of Computation, Discrete Mathematics and Graph Theory, Data Structures and Algorithms*

PROFESSIONAL EXPERIENCE

Bose Corporation

Shanghai, China

Assistant Embedded Software Engineer

Jul 2025 to Jul 2025

- Developed an ARM-based battery monitoring system, utilizing ADC for voltage reading and avoid exceeding the threshold by storing in Flash.
- Benchmarked system performance by comparing DMA and CPU-based methods for ADC data transfer, delivering data-driven recommendations for optimal architecture.
- Established a UART debugging framework to output critical system parameters and validate data integrity via SPI Flash polling.

Software Engineering Laboratory of Zhejiang University

Hangzhou, China

Backend Developer

Dec 2023 to Jan 2024

- Integrated ngSQL and Cypher queries for dynamic database interactions, using Nebula Graph and Starry Graph.
- Implemented database connectivity and query execution through Python and notebooks, focusing on database initialization, query optimization, and result encapsulation.
- Designed an abstract query interface for seamless database switching and streamlined graph database operations, including data visualization and analysis enhancements.
- Extended system capabilities to support s-cypher queries, culminating in the development of a comprehensive quick-start guide.

RESEARCH AND PROJECT EXPERIENCE

Trust in LLM-controlled Robotics: a Survey of Security Threats, Defenses and Challenges (Submitted to *IEEE Transactions on Robotics*)

Sydney, AUS

Research Assistant, Supervised by Prof. Huaming Chen from The University of Sydney

July 2025 to Present

- Conducted in-depth research on LLM-Controlled robotics to identify the major attack methods, including jailbreaking, backdoor attacks, and multi-modal prompt injection, and discover corresponding defense strategies.
- Introduced a novel taxonomy, capturing mechanisms across specification, validation, runtime enforcement, and assurance, along three axes, the safety lifecycle, the system layers, and the mechanism families.
- Pinpointed the "embodiment gap" as the root cause of unique physical security vulnerabilities in these systems and highlighted the inadequacy of traditional LLM safety measures in physical contexts.

A Survey of Domain Adaptation in Robotics Using Diffusion Models (Accepted by *3rd International Conference on Machine Learning and Automation 2025*)

Sydney, AUS

Research Assistant, Supervised by Prof. Cameron Riviere from Carnegie Mellon University

May 2025 to July 2025

- Performed systematic literature review on diffusion models and traditional domain adaptation techniques for robotics to analyze its effectiveness in bridging the sim-to-real gap.
- Illustrated the application and performance gains of diffusion models in robotic perception and manipulation through in-depth case studies of influential works like DiffuBox and 3D Diffusion Policy.
- Analyzed core deployment challenges, including the real-time performance bottleneck, safety verification, and the ongoing debate on model generalization versus memorization.

Preserving Digital Assets: Cryptocurrency Inheritance System

Sydney, AUS

Team Leader, Supervised by Yaoqi Huang from The University of Sydney

Aug 2024 to Nov 2024

- Architected asynchronous communication via RocketMQ for user registration, email, and wallet services.
- Implemented a distributed caching layer using Redis to significantly reduce database load by 50%, improving API response times by 50%.
- Deployed and continuously maintained the project on AWS.

OTHERS

Programming Languages: Java, Python, JavaScript, C, SQL, R

Packages: Pandas, PyTorch, TensorFlow, Scikit-learn

Networking & Protocols: TCP/IP, UDP, HTTP

Embedded System: STM32, Arduino, CAN/SPI protocols

Developer Tools: Git, GitHub, Gradle, Postman, IntelliJ, VS Code, Jira, Docker, Jupyter Notebook, AWS

Full Stack: Spring Boot, Django, HTML, JavaScript, Vue.js, React, Element-UI, Bootstrap, MyBatis, MySQL, REST API